Complex-Valued SAR Image Super-Resolution via Subaperture Learning and Fusion

Ganggang Dong, Yao Wang, Hongwei Liu, *Member, IEEE*, and Songlin Liu

***Abstract*— Synthetic aperture radar (SAR) imaging is a popular tool in many fields, while the contradiction between resolution and coverage is an open problem. The range resolution relies on the bandwidth of transmitted signals, while the azimuth resolution is dependent on the length of the synthesized aperture. They are related to radar configurations and are infeasible to be tuned freely. To solve the problem, a new SAR image super-resolution method is proposed in this article. It is composed of preimaging, feature learning, and reconstruction. In the first phase, the zero-padded technique is used to achieve spatial alignment. The result is then decomposed along the subapertures, forming the subaperture images and the subband images. They were delivered to the second phase for cross-domain learning and hierarchical fusion. The prelearning, encoder–decoder, and refinement are configured. The learned features are finally tuned for super-resolution reconstruction. On this basis, a new evaluation system was built to measure the performance. Both the supervised evaluation and the unsupervised evaluation are considered. An improvement of 3.78% is achieved for the amplitude components, while the phase accuracy is 4.62% better than the competitor.**

***Index Terms*— Deep learning, quality assessment, radar imaging, super-resolution, synthetic aperture radar (SAR).**

I. Introduction

ONE of the most important functions for radar was the high-resolution imaging [1], [2], [3]. It can be used to visualize the echo data and extract the important information using the downstream tasks, such as target detection [4] and recognition [5], [6], semantic segmentation [7], and registration [8]. The quality of upstream imaging, such as image resolution [9] and image auto-focus [10], played an important role in the downstream tasks. The primary factor to be considered is the image resolution. To delineate the effect of image resolution on target recognition performance, a set of examples are pursued on the MSTAR SAR dataset [11]. The experimental results are displayed in Table I. A set of experiments on SAR target recognition is pursued by images in different resolutions. A simple convolutional network (Wide-ResNet) was trained by SAR images in two different kinds of resolutions, including 0.3×0.3 and 0.6×0.6 m, respectively. The converged model was then tested by the images in these resolutions. As can be seen, the recognition accuracies were changed sharply. The best performance was 0.9385, compared to 0.1327 for the poorest accuracy. The results proved that the image resolution played an important role in the image interpretation. It was, therefore, much more valuable to achieve super-resolution reconstruction for the downstream tasks.

TABLE I
Effect of Image Resolution for Target Recognition. The Experimental Data Were from the MSTAR SAR Dataset [11], [12]

Train	0.3×0.3 Test	0.4×0.4 Test	0.5×0.5 Test	0.6×0.6 Test
0.3×0.3	0.9385	0.5393	0.3547	0.1327
0.6×0.6	0.4924	0.5279	0.7190	0.8200

A. Model-Driven Methods

For SAR imaging, the resolution along the range was dependent on the bandwidth of the transmitted signal. It can be improved by transmitting the modulated signal with a larger bandwidth. Likewise, the resolution along the azimuth relied on the length of the synthesized aperture. It can be improved by the longer observation accumulative time. However, the hardware configurations of radar systems are infeasible to be tuned freely in realistic applications. So, most of the previous works achieve the task of super-resolution by signal-processing techniques. The representative techniques can be summarized as follows.

1) Point Spread Function Deconvolution: Radar imaging can be viewed as the convolution of the echo signal with the system point spread function in theory. The image resolution can, therefore, be improved by the reverse operation by a finer point spread function (Fig. 1). Bose [13] presented a deconvolution algorithm for radar imaging of contiguous targets. A postprocessing block that works with the CLEAN algorithm [14] was used to discard the spurious peaks and consolidate the multiple small peaks corresponding to mislocated targets. Similar ideas can be found in [15] and [16]. However, the solution of deconvolution may be discontinuously dependent on the observed data. The slight

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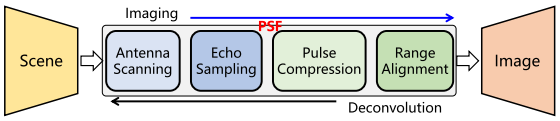


Fig. 1. Deconvolution for SAR super-resolution imaging.

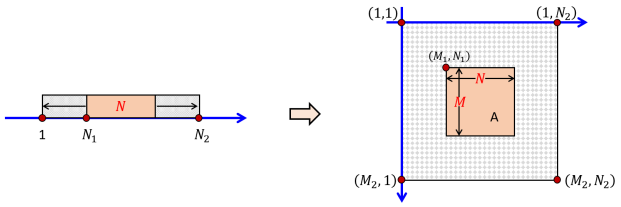


Fig. 2. Bandwidth extrapolation for SAR super-resolution imaging.

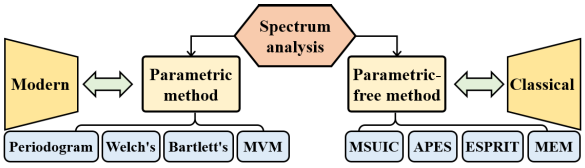


Fig. 3. Spectrum analysis methods for SAR super-resolution imaging.

disturbance in the echo signal may cause a great change in resolution.

2) Regularization Constraint: Radar imaging was actually a mathematical inversion problem to estimate the spatial position and the backscattering coefficient of the scattering source. It can be solved by regression with a regularization constraint. He et al. [17] enhanced the super-resolution imaging by a compressed sensing framework across the predefined dictionary. The low- and high-resolution subdictionary pair were jointly estimated for imaging. The similar work can be found in [18], [19], and [20].

3) Bandwidth Extrapolation: This family of methods estimated the data in outside bandwidth by the data of known band-limited signals in a certain bandwidth range. The mechanism is pictorially displayed in Fig. 2. It can be viewed as data extrapolation in the frequency domain. Xu and Narayanan [9] presented a refined iterative sidelobe apodization procedure for SAR image-resolution enhancement. Biondi [21] exported the spectrum damages that occur in the native data and focused on the survived spectrum information at a lower resolution.

4) Spectral Analysis: This family of methods produced the spectral in high resolution by the low-resolution spectral. Two kinds of paradigms, the parametric techniques and the parameter-free ones, were available (Fig. 3). The nonparametric method estimated the spectrum directly from the raw data [22]. Differently, the parametric method estimated the spectrum with some explicit models devised from the prior knowledge [23], [24], [25], such as APES, MUSIC, and ESPRIT [26], [27]. Ozsoy and Ergin [28] utilized the parametric modeling of the projected target reflectivity density function and tomographic reconstruction for super-resolution.

In these methods, an explicit model with a simple function was built first. The model parameters were estimated by the data available. They were used to predict the data absent, from which image super-resolution can be achieved. So, they suffer from some difficulties.

1) It is difficult to determine the optimal model. Take the auto-regressive model, for example, the performance was highly dependent on the order of the regression model. Yet, how to determine the optimal order was an open problem.
2) These methods were specific to certain data, or radar. The generalization ability was poor in the complex scenarios.
3) The prior imaging data have been ignored. The information losses were, therefore, produced.
4) These methods were related to system parameters. It cannot be implemented if the parameters are unavailable.

B. Data-Driven Methods

Recent studies have resorted to data-driven methods gradually. The high-level feature was learned from the input data for super-resolution reconstruction. There is no need to build an explicit physical model. The architectures popularly used can be summarized as the following routes.

1) Convolutional Model: This family of methods was typically composed of convolutional and pooling layers between the input and the output. Yoon et al. [29] first employed the convolutional network to learn an end-to-end mapping between the low/high-resolution images. They displayed state-of-the-art restoration quality and achieved fast speed for practical online usage. On this basis, the residual hierarchies can be applied, for example, the skip connection presented in the residual network was introduced for image super-resolution [30], [31].

2) Sequential Model: The most popularly used sequential model for image super-resolution was the Transformer and its variants, such as the Vision Transformer and the Swin Transformer. In these methods, the high-level features were learned from the input tokens. Chen et al. [32] proposed a fully Vision Transformer-based single-image super-resolution method. They demonstrate the advantages of a Transformer compared to the convolutional network. Xiao et al. [33] presented a top-k token selective Transformer for remote sensing image super-resolution. The interference of irreverent tokens for a more compact self-attention calculation was eliminated adaptively.

3) Generative Model: This family of methods achieved the task of super-resolution by the generative models, such as the generative adversarial network (GAN), the variable autoencoder (VAE), and the diffusion model. Ledig et al. [34] presented a generative adversarial network for image super-resolution (SRGAN). A perceptual loss function that consists of an adversarial loss and a content loss was developed for network training. Wang et al. [35] achieved the super-resolution resolution by a single-step generation model. A deterministic sampling process from the most recent state-of-the-art method for accelerating diffusion-based models was developed. Xiao et al. [36] employed the diffusion probabilistic model for remote sensing image super-resolution.

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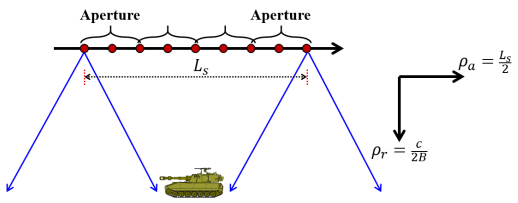


Fig. 4. SAR imaging, where ρ_r and ρ_a denote the resolution along the range and the azimuth, respectively, and L_s is the length of the virtually synthesized antenna.

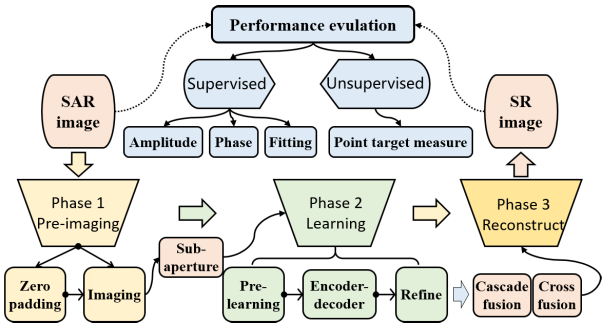


Fig. 5. Overall framework of this article.

An efficient activation network was developed to boost the noise prediction performance by simplified channel attention and simple gate operation.

Though great performance was achieved in the preceding works, some issues should be considered further.

1) These methods were developed for the real-valued vision images. SAR images were formed by the coherent imaging mechanism. The IQ quadrature sampling made the image complex-valued. How to handle the amplitude and the phase simultaneously was challenging.
2) Though some previous works were devoted to SAR image super-resolution, only the amplitude components were reconstructed. The phase information and the imaging mechanism were ignored directly.
3) The general vision metrics were employed to evaluate the reconstruction accuracy of amplitude. How to measure the performance of SAR image super-resolution comprehensively should be studied further.

C. Solution

To solve those problems, a new subaperture learning and hierarchical fusion method was presented in this article. It was used to achieve complex-valued SAR image super-resolution tasks. The proposed method was composed of three phases, preimaging via zero padding, feature learning via encoder–decoder architecture, and image reconstruction. In the first phase, the zero padding in the frequency domain was employed to form an initialized image. The input image can be aligned roughly on the spatial scale. The preimaging result was then decomposed along the subaperture and the subband [37], [38], forming the subaperture images and the subband images. The mechanism of synthetic aperture radar (SAR) imaging was delineated in Fig. 4.

The decomposition results were then delivered to the second phase, where three sequential modules—prelearning, encoder–decoder, and feature refinement—were configured. The hierarchical features were learned from the subaperture images by an encoder–decoder. These features were then combined by two kinds of fusion techniques: cascade fusion and cross-fusion. The results were finally refined and tuned to produce the super-resolution result. On this basis, an evaluation system for complex-valued SAR image super-resolution was established. Both the supervised and unsupervised evaluations were considered. Five kinds of metrics—amplitude accuracy, phase congruence, fitting precision, information content, and the point target measurement—were used to quantify the performance. The problems presented above can, therefore, be circumvented.

1) The complex-valued SAR image was regarded as a high-dimensional tensor. It was fed into the proposed subaperture learning and hierarchical framework for super-resolution reconstruction. So, the amplitude and the phase components were jointly exploited.
2) The zero padding technique was employed to form the initialized image. It was then decomposed across the subaperture. The high-level semantic features in different scales can be learned. The imaging mechanism was therefore exploited in the framework of deep learning.
3) A new evaluation system was presented. Both the supervised evaluation and the unsupervised assessment were considered. The super-resolution performance can, therefore, be evaluated comprehensively.

The details are pictorially shown in Fig. 5.

In the end, multiple rounds of experiments were performed. The experimental results throw light on the superiority of the proposed method over the standard and other recent techniques. It serves to generate images with a better reconstruction quality. Likewise, the established evaluation system could measure the quality of SAR images fairly and comprehensively.

The remainder of this article was organized as follows. Section II reviews the signal model of SAR imaging. Section III presented the proposed method, where the three phases for super-resolution imaging were detailed. The developed evaluation system for radar imaging was described in Section IV. Five kinds of metrics were presented to measure the imaging quality. Section V verified the proposed method and the developed evaluation system by some quantitative experiments. Section VI concluded this article and outlined the future research.

II. Signal Model

According to the Heisenberg principle, the resolution along the time axis and the frequency axis were contradictory in radar technology. To obtain a better signal-to-noise ratio, the signals with large time-bandwidth products were transmitted. The high resolution along the range and the azimuth can be then achieved simultaneously. Take the linear frequency modulation signal (LFM), for example, the transmitted signal

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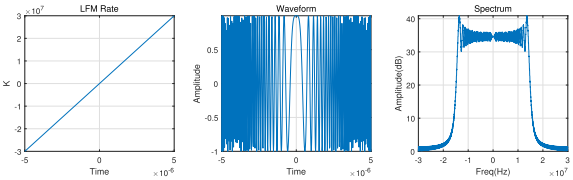


Fig. 6. Illustration of the LFM signal.

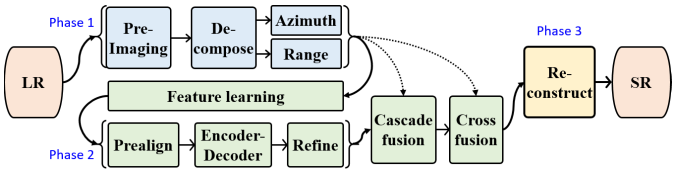


Fig. 7. Outline of the proposed SAR super-resolution architecture.

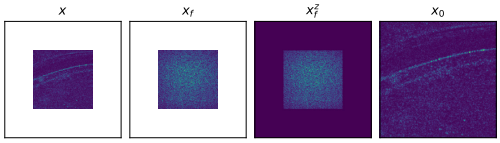


Fig. 8. Illustration of the proposed zero-padding imaging technique. (Left) Two figures displayed the SAR image and its spectrum in poor resolution (amplitude), while (right) two figures showed the preimaging result (amplitude).

was expressed as

$$s(\tau) = \text{rect}\left(\frac{\tau}{T_p}\right) \cdot \exp(j2\pi f_c \tau + j\pi K \tau^2) \tag{1}$$

where rect(·) denotes the window function to form the pulse signals, f_c is the carrier frequency, $K = (B/T_p)$ is the LFM rate, T_p is the pulse repetition period, and B is the bandwidth of the LFM signal. A simple exemplar of the LFM is displayed in Fig. 6. The echo signal from the imaging scenes can be viewed as the convolution of the ground reflection coefficients with the transmitted signal

$$s_r(\tau) = g_r(\tau) \otimes s(\tau) \tag{2}$$

where $g_r = \sigma\delta(\tau - \tau')$ and $\tau' = (2R_a/c)$ is the time delay. The received signal can then be simply written as

$$s_r(\tau) = \sigma \cdot \text{rect}\left(\frac{\tau - \tau'}{T_p}\right) \cdot \exp\left(j2\pi f_c(\tau - \tau') + j\pi K(\tau - \tau')^2 + \phi\right) \tag{3}$$

where ϕ is the change of signal phase resulting from the ground surface scattering. Suppose the antenna pattern was theoretically approximated as the sinc function. The final echo signal can be modeled as

$$s_r(\tau, \eta) = \sigma \cdot \omega_r(\tau - \tau') \cdot \omega_a(\eta - \eta') \times \exp\left(j2\pi f_c(\tau - \tau) + j\pi K(\tau - \tau)^2 + \phi\right) \tag{4}$$

where $\omega_r(\tau) = \text{rect}(\tau/T_p)$ is the envelop function along the range and $\omega_a(\eta)$ is the envelop function along the azimuth. The resolution along the range and the azimuth is

$$\rho_r = \frac{c}{2B}, \quad \rho_a = \frac{L_s}{2} \tag{5}$$

where B is the signal bandwidth and L_s is the length of the synthetic antenna aperture. The illustration of SAR image resolution can be found in Fig. 4.

III. Proposed Method

In this article, the task of SAR image super-resolution is achieved by cross-domain learning and hierarchical fusion. The learning paradigm is outlined in Fig. 7. It is composed of three sequential phases: the preimaging, the feature alignment, and the reconstruction.

A. Phase 1—Preimaging

The first step for the proposed SAR super-resolution method is to align the spatial scale. For the real-valued vision image, it is usually achieved by the linear interpolation strategy, such as the nearest neighbor interpolation, the bilinear interpolation, and the cubic interpolation [39]. Since the SAR image is complex-valued, these techniques were not applicable. In this article, the zero-padding imaging technique is presented to achieve the scale alignment. The results are then decomposed along the range and the azimuth axes, forming the subaperture images and the subband images, respectively.

1) Preimaging: The complex-valued SAR image is first recast into the frequency domain, forming the single periodic frequencies by spectrum shifting. The scale alignment is then achieved by filling the zeros values along the dc components direction. The aligned image can be then produced by re-imaging the padded frequencies. The overall process can be expressed as

$$\mathbf{x} \overset{\mathcal{F}(\cdot)}{\rightarrow} \mathbf{x}_f \overset{\mathcal{Z}(\cdot)}{\rightarrow} \mathbf{x}_f^z \overset{\mathcal{F}^{-1}(\cdot)}{\rightarrow} \mathbf{x}_0 \tag{6}$$

where $\mathcal{F}$ and $\mathcal{Z}$ denote the Fourier transform and the zero-padding operations, respectively, and $\mathbf{x}_0$ is the initialized aligned result. To further demonstrate the proposed preimaging technique, a set of exemplars is shown in Fig. 8.

Compared to the general interpolation skills, the proposed preimaging has several characteristics.

1) The radar imaging is actually a series of signal processing between the time domain and the frequency domain. So, the zero padding in one domain could promote the resolution in another domain.
2) Different from the preceding works, a finer initialization was generated by the zero padding imaging in this article. The spatial scale of low-resolution images can be aligned well with the fine-resolution.
3) Both the phase and the amplitude components of the SAR image can be preserved in the proposed framework.

The preimaging result $\mathbf{x}_0$ (initialization) is used for subaperture decomposition.

2) Subaperture Decomposition: The resolution of real aperture imaging was dependent on the range of the target to radar, as well as the size of the antenna

$$\rho_r = R \cdot \theta_{3\,\text{dB}} = R \cdot \frac{\lambda}{\ell} \tag{7}$$

where R is the range of the target to radar and D is the length of the radar antenna. So, it suffers some difficulties in practice.

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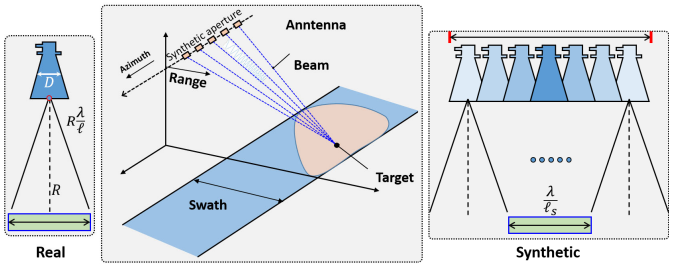


Fig. 9. Illustration of real and SAR imaging mechanism, ℓ is the length of the real aperture antenna, while ℓ_s is the length of the synthesized antenna.

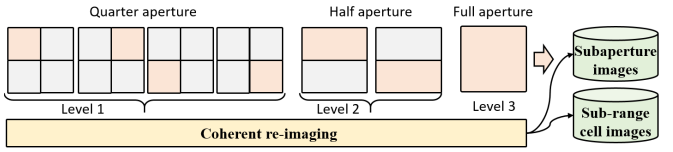


Fig. 10. Illustration of SAR subaperture decomposition imaging.

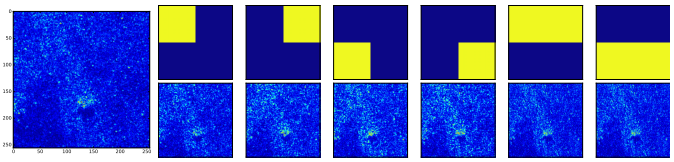


Fig. 11. Exemplars on SAR subaperture decomposition imaging.

The mechanism was delineated in the left subfigure of Fig. 9. Contrarily, the resolution along the azimuth and the range were improved by synthetic aperture imaging. The motion of the radar antenna along the azimuth was synthesized in a coherent process interval (CPI). A sequence of acquisitions from a shorter antenna can be then combined to simulate a much larger antenna, thus providing higher resolution data [3]. The mechanism was simply illustrated Fig. 9. The resolution along the range and the azimuth was

$$\{\rho_r, \rho_a\} \rightleftharpoons \left\{\frac{c}{2B}, \frac{\ell_s}{2}\right\} \tag{8}$$

where B is the bandwidth and ℓ_s is the length synthesized antenna. So, the range resolution was dependent on the signal bandwidth only, while the azimuth resolution was dependent on the length of the synthesized antenna.

To learn the more finer features in different scales, this article proposes to decompose the full-aperture image (preimaging result) along the range and the azimuth, respectively. Here, two decomposition levels are considered. It is delineated in Fig. 10. In the first level, four subimages are obtained by re-imaging on a quarter of the original frequency. Four series of images can be produced accordingly. In the second level, two subimages are obtained by re-imaging half of the original frequency along the azimuth axis. Two series of images can be formed likewise. To delineate the proposed decomposition, a set of exemplars is shown in Fig. 11.

The decomposition results are then delivered to the second phase for feature learning and information fusion.

B. Phase 2—Feature Learning

The second step of the proposed method is feature alignment by cross-domain learning and hierarchical fusion. It is composed of prelearning, encoder–decoder, and refinement. Two kinds of information fusion, cross fusion and cascade fusion, are developed to enhance learning effectiveness. The overall framework of feature alignment is displayed in Fig. 12.

1) Prelearning: The prelearning module inputs the subaperture images and subband images and outputs the initialized learned representations. It consists of the convolution blocks mainly. In each block, the attention learning mechanism in conjunction with the convolution operations is implemented [31]. The attention mechanism includes squeeze, excitation, and scaling operations. They are combined to produce the initialized feature maps [40]. The framework of the prelearning module is displayed in Fig. 13.

Given the input feature $\mathbf{F}^{(\text{in})}$, the average global pooling is first applied $\phi_p(\mathbf{F}^{(\text{in})})$. The coefficients are then fed into two hierarchical convolution layers. In the former convolution, ReLU is used to achieve activation, while the sigmoid function and the gate mechanism are applied in the latter convolution. The attention mechanism map is calculated by

$$\mathbf{F}^{(\text{att})} = \sigma\left(\phi_{\text{conv}}\left(\delta\left(\phi_{\text{conv}}\left(\phi_p\left(\mathbf{F}^{(\text{in})}\right)\right)\right)\right)\right) \tag{9}$$

where ϕ_{conv} denote the convolutions and $\sigma(\cdot)$ and $\delta(\cdot)$ are the sigmoid and the ReLU function, respectively. The output features are then obtained by weighting the input features with the attention map

$$\mathbf{F}_{\text{pre}}^{(\text{out})} = \oplus\left(\otimes\left(\mathbf{F}^{(\text{att})}, \Omega\right), \mathbf{F}^{(\text{in})}\right) \tag{10}$$

where Ω denotes the attention weights. Several convolution blocks are then cascaded to form the prelearning module. The input is the scale-alignment image $\mathbf{x}_0$, while the output is the initialized learned features $\mathbf{F}_{\text{pre}}^{(\text{out})}$. They are further delivered to the deep alignment module.

2) Encoder–Decoder: The encoder–decoder module inputs the initialized representations $\mathbf{F}_{\text{pre}}^{(\text{out})}$ and outputs the encoded representations $\mathbf{F}_{\text{deep}}^{(\text{out})}$. The semantic information in different scales can, therefore, be extracted.

1) During the encoding phase, the initialized features $\mathbf{F}_{\text{pre}}^{(\text{out})}$ are downsampled to get the semantic information in the poorer scales. In each scale, the attention mechanism conjunction with convolution is used to learn the features. It is similar to the prelearning

$$\mathbf{F}_{\text{deep}}^{(\text{en})} = \phi_{\text{encoder}}\left(\mathbf{F}_{\text{pre}}^{(\text{out})}\right). \tag{11}$$

2) During the decoding phase, the feature maps are reconstructed in the finer scales by the upsampling followed by convolution operations

$$\mathbf{F}_{\text{deep}}^{(\text{de})} = \phi_{\text{decoder}}\left(\mathbf{F}_{\text{deep}}^{(\text{en})}\right) \tag{12}$$

where $\phi_{\text{encoder}}, \phi_{\text{decoder}}$ denote the abstract mapping of the encoder and the decoder. The encoder–decoder framework was delineated in Fig. 14. Likewise, the features in each scale are tied in skip connection to prevent the disappearance of the gradient.

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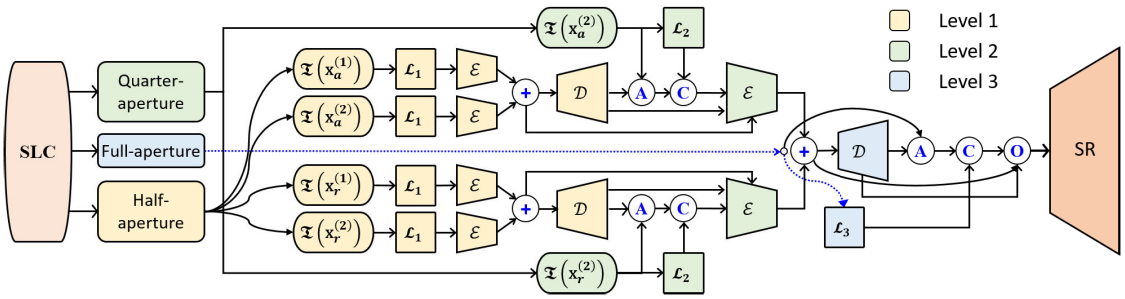


Fig. 12. Proposed subaperture learning framework, where $\mathcal{I}$ denotes the re-imaging, $\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3$ is the three-level fusion, and $\mathcal{D}$ and $\mathcal{E}$ are the decoder and the encoder. The operations of addition, concatenation, and full-resolution convolution are noted as A, C, and O, respectively.

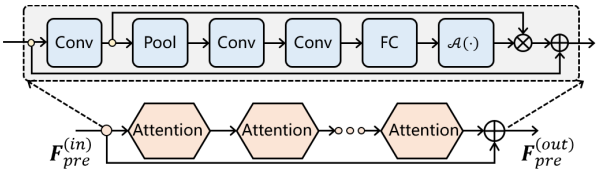


Fig. 13. Prelearning module, where $\mathcal{A}$ denotes the activation function.

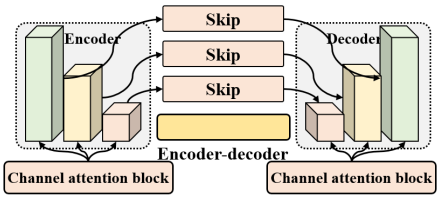


Fig. 14. Encoder–decoder framework.

3) Refinement: The refinement module inputs the encoded features and outputs the finer representations. It consists of attention learning supervised by the full-resolution image. Different from the common regression tasks, the full-resolution image is introduced to boost the alignment accuracy [41]. Only the informative features can be delivered to the subsequent phase, while the redundant information is filtered out.

Given the input features $\mathbf{F}^{(\text{in})}$, the residual image is first calculated

$$\begin{cases} \mathbf{x}_{\text{diff}} = \phi_{\text{conv}}^{@1\times1}\left(\mathbf{F}_{\text{post}}^{(\text{in})}\right) \\ \mathbf{x}_{\text{ref}} = \ominus(\mathbf{x}_{\text{diff}}, \mathbf{x}_{\text{hr}}) \end{cases} \tag{13}$$

where $\mathbf{x}_{\text{hr}}$ is the ground truth. It is then added to the original image to reconstruct the image in finer resolution

$$\mathbf{x}_r = \oplus(\mathbf{x}_{\text{ref}}, \mathbf{x}_0). \tag{14}$$

The pixel-wise convolutions (with 1×1 kernel) followed by the activation function are then applied to produce the pixel-wise attention masks

$$\mathbf{F}^{(\text{att})} = \sigma\left(\phi_{\text{conv}}^{1\times1}(\mathbf{x}_r)\right) \tag{15}$$

where $\phi_{\text{conv}}^{1\times1}(\cdot)$ denotes the pixel-wise convolutions and $\sigma(\cdot)$ is the sigmoid gate operation. The features are finally obtained by the residual shortcut connection followed by the attention weighting

$$\mathbf{F}_{\text{post}}^{(\text{out})} = \oplus\left(\otimes\left(\mathbf{F}^{(\text{att})}, \phi_{\text{conv}}\left(\mathbf{F}^{(\text{in})}\right)\right), \mathbf{F}^{(\text{in})}\right). \tag{16}$$

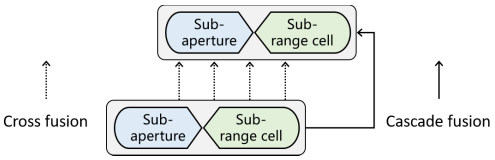


Fig. 15. Illustration of proposed hierarchical fusion for SAR super-resolution.

The learned features are then delivered to the feature refinement phase.

C. Hierarchical Fusion

The learned features are then combined by two kinds of information fusion, including cascade fusion and cross fusion. The cascade fusion refers to connecting the features in the different stages from tail to head. Contrarily, cross-fusion means concatenating the decoding features of the previous stage with the encoding features of the latter stage. The proposed hierarchical fusion technique is pictorially shown in Fig. 15.

1) Cascade Fusion: The cascade fusion refers to connecting two-stage hierarchies in order. To extract more semantic information, this article decomposes the full-aperture image along the range and the azimuth, respectively. The resulting subimages are then fed into the individual modules for feature fusion.

1) *Information Fusion Over the Azimuth Axis:* In the first stage, the original full-aperture image was decomposed along the azimuth direction. A set of subaperture images were generated by imaging on a part of radar measurements [42], [43]. They were fed into the individual convolution modules. Take two subaperture decompositions, for example, the feature fusion can be expressed as

$$\mathbf{x}_0 \xrightarrow{\mathcal{F}} \mathbf{x}_f \Rightarrow \left\{ \begin{matrix} \mathbf{x}_{f,1}^{(a)} \xrightarrow{\phi_{\text{conv}}} \mathbf{F}_l^{(a)} \\ \mathbf{x}_{f,2}^{(a)} \xrightarrow{\phi_{\text{conv}}} \mathbf{F}_r^{(a)} \end{matrix} \right\} \oplus \Rightarrow \mathbf{F}^{(a)} \tag{17}$$

where $\mathbf{x}_f$ denote the frequency spectrum of $\mathbf{x}_0$ and $\mathbf{x}_{f,1}^{(a)}, \mathbf{x}_{f,2}^{(a)}$ are the two subaperture images. The learned features are then combined to integrate the information extracted from the subaperture images.

2) *Information Fusion Over the Range Axis:* In the second stage, the full-aperture images are decomposed along the range direction similarly. A set of subrange-cell images

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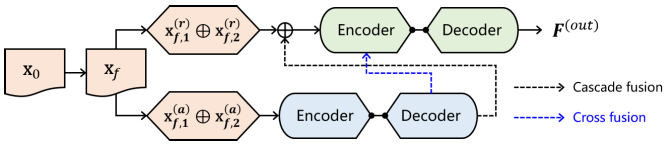


Fig. 16. Diagram of information fusion in feature alignment.

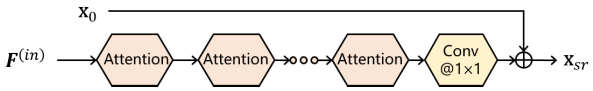


Fig. 17. Feature refinement in the third phase.

are produced by imaging on a part of measurements accordingly [44]. They are fed into the individual convolution modules. Take two subrange-cell images, for example, the fusion can be expressed as

$$\mathbf{x}_0 \xrightarrow{\mathcal{F}} \mathbf{x}_f \Rightarrow \left\{ \begin{array}{l} \mathbf{x}_{f,1}^{(r)} \xrightarrow{\phi_{\text{conv}}} \mathbf{F}_t^{(r)} \\ \mathbf{x}_{f,2}^{(r)} \xrightarrow{\phi_{\text{conv}}} \mathbf{F}_b^{(r)} \end{array} \right\} \oplus \Rightarrow \mathbf{F}^{(r)} \tag{18}$$

where $\mathbf{x}_{f,1}^{(r)}, \mathbf{x}_{f,2}^{(r)}$ are the two range-cell images. The learned features are then combined to integrate the information extracted from the subrange-cell images.

The learned features from the subaperture images and the subrange-cell images are then combined

$$\mathcal{F}_{\text{cascade}} = \oplus(\mathbf{F}^{(a)} + \mathbf{F}^{(r)}). \tag{19}$$

2) Cross-Fusion: In the cascade fusion, the information was delivered from the tail of the previous stage to the head of the latter stage. Differently, cross-fusion refers to combining the decoder components of the previous stage with the encoder components of the latter stage

$$\mathcal{F}_{\text{cross}} = \oplus\left(\phi_{\text{decoder}}^1\left(\mathbf{F}_{\text{deep}}^{(\text{en})}\right), \phi_{\text{encoder}}^2\left(\mathbf{F}_{\text{pre}}^{(\text{out})}\right)\right) \tag{20}$$

where ϕ_{decoder}^1 and ϕ_{encoder}^2 denote the decoder mapping of the previous stage and the encoder mapping of the latter stage, respectively.

3) Overall Architecture: The two levels of feature fusion are illustrated in Fig. 16. The cascade fusion is highlighted in a dotted arrow in black, while the cross fusion is shown in blue.

The learned features resulting from feature alignment were then delivered to the subsequent feature refinement phase.

D. Phase 3—Reconstruction

On the foundation of feature alignment, the learned features are further refined by the tail hierarchy. It adopts the channel attention mechanism similar to the prelearning in the second phase. The box diagram of feature refinement is displayed in Fig. 17, where the details on the attention block can be found in Fig. 13.

Given the input features $\mathbf{F}_{\text{pre},3}^{\text{out}}$, the result of super-resolution imaging can be expressed as

$$\mathbf{x}_s = \otimes\left(\mathbf{x}_0, \Phi_r\left(\mathbf{F}_{\text{pre},3}^{\text{out}}\right)\right) \tag{21}$$

where $\Phi_r(\cdot)$ denotes the feature refinement mapping.

E. Training Strategy

The loss function was used to measure the predicted errors, by which the gradient descent algorithm can be implemented. The optimal weights of the proposed deep architecture can be then estimated. So, the loss function played an important role in the training procedure. Different from the vision tasks, the radar images were complex-valued. The amplitude and the phase components were available simultaneously. In this article, both of them were considered in the calculation of loss measurements.

1) Amplitude Loss: The amplitude components can be visualized similarly to the vision image. So, the common loss functions can be used to measure the prediction error, such as the mean square error (ℓ_2 norm) and the mean absolute error (ℓ_1 norm). Both of them have their own advantages and limitations. In this article, a compromise function, the pseudo-Huber loss is used to measure the alignment accuracy

$$\ell_h(\mathbf{x}_s, \mathbf{x}_h) = \sqrt{(\psi_e(\mathbf{x}_s, \mathbf{x}_h))^\alpha + \epsilon} \tag{22}$$

where $\psi_e(\cdot,\cdot)$ is the function to measure the alignment error between the predicted image and the ground truth and ϵ is a small constant to prevent the disappearance of the gradient. It is capable of balancing the tradeoff between the reconstruction accuracy and the robustness.

2) Phase Loss: The phase values of the SAR image are ranged from $-\pi$ to π. It is expected that the phase error can be as small as possible. So, this article quantified the phase error by the MAE loss

$$\ell_p(\mathbf{x}_s, \mathbf{x}_h) = \text{MAE}(\mathbf{x}_s), \phi(\mathbf{x}_h)) \tag{23}$$

where $\phi(\cdot)$ denotes the function to calculate the phase of a complex-valued image.

The overall loss function is defined as a linear combination of the amplitude loss and the phase loss

$$\ell(\mathbf{x}_s, \mathbf{x}_h) = \ell_h(\mathbf{x}_s, \mathbf{x}_h) + \lambda \cdot \ell_p(\mathbf{x}_s, \mathbf{x}_h) \tag{24}$$

where λ is the free parameter to control the relative importance of the two loss terms.

IV. Evaluation System

Different from the vision image, SAR images result from the intermediate-frequency quadrature sampling. The measurement results were complex-valued. So how to measure the super-resolution performed is still an open problem. In the preceding works, the general metrics for vision images are employed directly. They are yet not effective in complex-valued SAR images. To solve the problem, a new systematic assessment strategy is proposed in this article. The supervised evaluation as well as the unsupervised assessment are considered. The super-resolution results are assessed from five different perspectives, including the amplitude accuracy, the phase congruence, the fitting precision, the information content, and the point target imaging measurement. The proposed evaluation system for SAR super-resolution is illustrated in Fig. 18. Next, denote by $\mathbf{x}_{\text{hr}}, \mathbf{x}_{\text{sr}}$ as the original high-resolution image and the super-resolution result, respectively.

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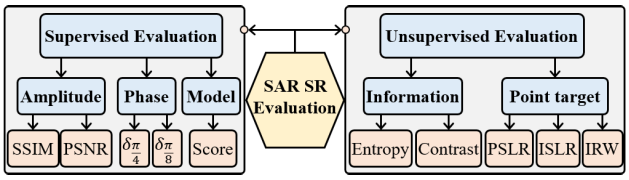


Fig. 18. Proposed assessment system for SAR image super-resolution.

A. Supervised Evaluation

The supervised evaluation refers to measuring the super-resolution performance in the course of the original full-resolution image. The presented assessment metrics include amplitude accuracy, phase congruence, and fitting precision.

1) Amplitude Accuracy: In practical applications, the single-look complex image can be visualized by its amplitude/envelop components. The reconstruction accuracy of amplitude was assessed by the structural similarity (SSIM) and the peak signal-to-noise ratio (PSNR).

1) SSIM is defined as the ratio of some mixed statistics

$$\phi_{\text{ssim}}(\mathbf{x}_{\text{hr}}, \mathbf{x}_{\text{sr}}) = \frac{(2\mu_t\mu_s + \epsilon_1)(2\sigma_{\text{ts}} + \epsilon_2)}{\left(\mu_t^2 + \mu_s^2 + \epsilon_1\right)\left(\sigma_t^2 + \sigma_s^2 + \epsilon_2\right)} \tag{25}$$

where $\mu_t, \mu_s, \sigma_t^2, \sigma_s^2$ is the mean and the variance of the truth image and the super-resolution result and ϵ_1, ϵ_2 are two small constants to avoid the division by zero. The values of SSIM are range from -1 to 1. The higher the value, the better the reconstructed amplitude.

2) PSNR is the ratio of signal peak energy to the average energy of noise

$$\phi_{\text{psnr}}(\mathbf{x}_{\text{hr}}, \mathbf{x}_{\text{sr}}) = 20 \cdot \log_{10}\left[\frac{v_{\max}}{\text{mse}(\mathbf{x}_{\text{hr}}, \mathbf{x}_{\text{sr}})}\right] \tag{26}$$

where $v_{\max}$ is the peak energy of signal and $\text{mse}(\cdot,\cdot)$ denotes the mean square error. The larger the value, the better the reconstructed amplitude.

2) Phase Congruence: The vision images were formed by the aggregation of lights. Differently, the radar images were obtained by the electromagnetic measurements of the echo signal. The formed images were complex-valued by the intermediate frequency quadrature sampling strategy. So, how to measure the quality of the reconstructed phase was a unique problem for radar images. In the preceding works, the evaluation of phase is usually ignored. In this article, a new metric, the phase congruence, was developed. The absolute errors of the super-resolution results and the truth image are first calculated

$$\mathbf{E}(\mathbf{x}_{\text{hr}}, \mathbf{x}_{\text{sr}}) = \text{mae}(\psi(\mathbf{x}_{\text{sr}}), \psi(\mathbf{x}_{\text{hr}})) \tag{27}$$

where $\psi(\cdot)$ denotes the function to calculate the phase of a complex-valued image. The statistical histogram of phase errors is then quantized. Phase congruence is defined as the ratio of errors less than a certain threshold τ

$$\delta_\tau(\mathbf{x}_{\text{hr}}, \mathbf{x}_{\text{sr}}) = \Omega(\mathbf{E} \le \tau) \tag{28}$$

where $\Omega(\cdot)$ denotes the count function. So, the metric is related to the constant τ. Take $\delta_{(\pi/8)}, \delta_{(\pi/4)}$, for example, the metrics

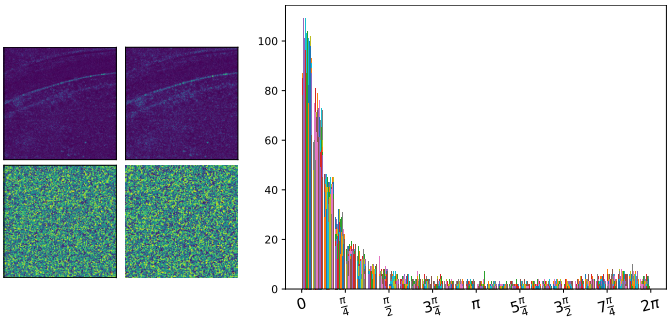


Fig. 19. Illustration of the error histogram for phase congruence. (Left) Pair of radar images (amplitude and phase), while (right) histogram is displayed.

can be expressed as

$$\delta_{\frac{\pi}{8}}/\delta_{\frac{\pi}{4}}(\mathbf{x}_{\text{hr}}, \mathbf{x}_{\text{sr}}) = \Omega\left(\mathbf{E} \le \frac{\pi}{8}\Big/\frac{\pi}{4}\right). \tag{29}$$

The bigger the metric δ, the finer the reconstructed phase, and hence the better the super-resolution performance. To delineate the presented metric, a pair of images were displayed in Fig. 19. The measurements are shown on the right.

One major contribution of this article is to present a new metric to measure phase congruency. Here, it should be noted that the presented metric can also be applied to other imaging tasks, such as image refocus, jamming suppression, and sidelobe suppression.

3) Fitting Precision: The task of SAR image super-resolution can be modeled as a regression problem, where the image in low resolution $\mathbf{x}_{\text{lr}}$, the super-resolution result $\mathbf{x}_{\text{sr}}$ are regarded as the independent variable, and the dependent variable, respectively. The problem is solved by a three-phase deep architecture. To measure the fitting performance of the model, the precision is then developed

$$\kappa(\mathbf{x}_{\text{hr}}, \mathbf{x}_{\text{sr}}) = 1 - \frac{\text{mse}(\mathbf{x}_{\text{hr}}, \mathbf{x}_{\text{sr}})}{\text{var}(\mathbf{x}_{\text{hr}})} \tag{30}$$

where $\text{mse}(\cdot,\cdot)$ denotes the mean square error function and $\text{var}(\cdot)$ is to calculate the variance. The closer the precision rate was to 1, the better the fitting performance of the regression model.

B. Unsupervised Evaluations

Different from the supervised evaluation, the unsupervised assessment does not need the reference image. The metric can be calculated directly from the image available. The presented assessment metrics include the information content and the point target imaging. The metrics of images in low-resolution, super-resolution, and full-resolution can be calculated, respectively.

1) Information Content: The regression problem can also be viewed as the optimal estimation of random signals from the perspective of information theory. The optimal criteria include the minimum entropy and the maximum contrast. In this article, these criteria are employed to measure the super-resolution performance. Two metrics, the image entropy and the image contrast are presented.

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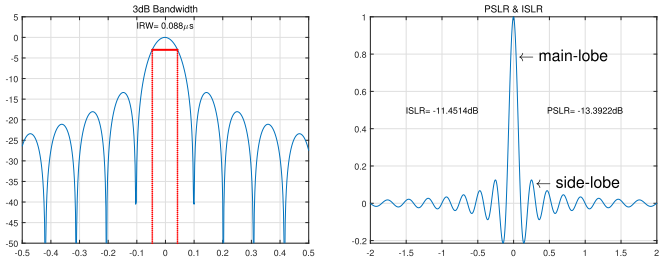


Fig. 20. Calculation of IRW, PSLR, and ISLR.

1) The information entropy is used to measure the overall characteristics of the signal on average. It is defined as

$$\phi_e(\mathbf{x}) = -\sum_x p(x)\log_2(p(x)) \tag{31}$$

where $p(x)$ is the statistical distribution of the random variable $\mathbf{x}$. Here, it is used to assess the quality of a super-resolution image. The smaller the entropy, the more stable the super-resolution result.

2) The image contrast is defined as

$$\phi_c(\mathbf{x}) = \frac{\sqrt{\mathbb{E}(|\mathbf{x}|^2 - \mathbb{E}(|\mathbf{x}|^2))}}{\mathbb{E}(|\mathbf{x}|^2)} \tag{32}$$

where $\mathbb{E}$ is the mathematic expectation. The higher the contrast, the more information the super-resolution result.

2) Point Target Measurement: For SAR imaging techniques, the most representative method to measure the image quality was the point target metrics. The metrics include the impulse response width (IRW), the peak sidelobe ratio (PSLR), and the integrated sidelobe ratio (ISLR) [45].

1) IRW is actually the 3-dB width of the mainlobe of the signal, $\phi_{3\,\mathrm{dB}}(\mathbf{r})$, where $\mathbf{r}$ is the profile along the range or the azimuth. A simple example is displayed in Fig. 20.
2) PSLR is defined as the ratio of the first (or maximum) sidelobe peak to the mainlobe peak

$$\delta_{\mathrm{pslr}} = 20\cdot\log_{10}\left[\frac{\psi_p^{(1)}(\mathbf{r})}{\psi_p^{(0)}(\mathbf{r})}\right] \tag{33}$$

where $\psi_p^{(0)}(\mathbf{r})$, $\psi_p^{(1)}(\mathbf{r})$ denote the peak value of mainlobe and first sidelobe.

3) ISLR is defined as the energy of the sidelobe to the right of the mainlobe divided by the one of the right half of the mainlobe

$$\delta_{\mathrm{islr}} = 20\cdot\log_{10}\left[\frac{\psi_r^{(1)}(\mathbf{r})}{\psi_r^{(0)}(\mathbf{r})}\right] \tag{34}$$

where $\psi_r^{(0)}(\mathbf{r})$, $\psi_r^{(1)}(\mathbf{r})$ denote the energy of right half of the mainlobe and the sidelobe.

For each image patch, IRW, PSLR, and ISLR along the azimuth and the range can be calculated. The smaller the metrics, the better the imaging resolution. A set of exemplars is displayed in Fig. 20.

However, the point target model was no longer invalid in the realistic scenarios. The metrics were, therefore, infeasible to be calculated directly. In this article, the 64×64 pixels around a certain peak are first cropped from the original image. The cropped patch was then interpolated sometimes by the sinc function [46]. The metrics were finally calculated from the interpolated image.

TABLE II
PARAMETERS FOR RADAR IMAGING

	azimuth	range
Image dimension	3000	1754
GRP pixel	1500	877
Scale factors (m/p)	0.10003634	0.10003634
Resolution (m)	0.12000185	0.12002461
Oversample ratios	1.19958255	1.19981015
Zero-pad ratios	1.01060939	1.01036012
Image size (m)	300	175

V. EXPERIMENTS AND DISCUSSIONS

This article achieved the task of complex-valued SAR image super-resolution by a three-phase deep architecture. On this basis, an evaluation system was established to assess the super-resolution performance. To verify the proposed method and the evaluation system, multiple rounds of comparative experiments were performed. The results demonstrated the resolution can be enhanced by the proposed method. Likewise, the presented evaluation system was effective in measuring the super-resolution quality.

A. Experimental Setups

The radar images used in this article were collected by Sandia National Laboratory on 14th February 2006. They were publicly released on the official website. The collection consisted of two flights namely "am" and "pm." Each of these flights contains 32 spotlight SAR images near Albuquerque, New Mexico. The flight was further broken down into eight flight passes that included four images per pass. The details on system parameters were displayed in Table II.

To make the computational cost acceptable, the original images were divided into 256×256 pixels, forming a total number of 3584 patches. Some exemplars of the patches are displayed in Fig. 21. The patches available for training and testing are set as 3084 and 500 in total, respectively. The evaluation metrics are obtained by averaging all the testing samples. The complex-valued patches ($\mathbf{x}$) were first recast into the frequency domain. The inverse operations of zero-padding and windowing were applied. The frequency components of full resolution image were then reduced by a factor, such as $2^{-1}, 2^{-2}$. They were used to generate the low-resolution images via re-imaging. The pairs of images can be then used to train the SAR super-resolution architecture.

B. Verifications of Preimaging

In the preceding works, the linear interpolation techniques were used for spatial alignment. Yet, they were not effective in complex-valued SAR images. To solve the problem, this article proposes the preimaging for image alignment. We recast

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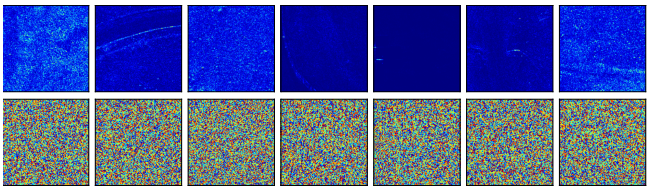
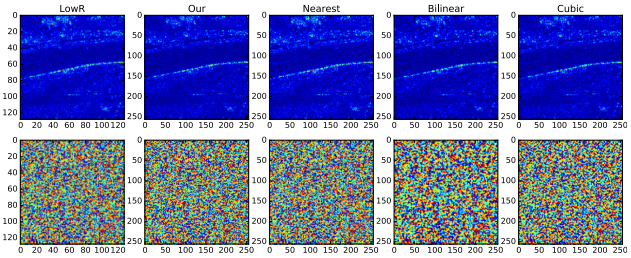


Fig. 21. Exemplars on SAR image patches: (Top) amplitude and its (bottom) corresponding phase is displayed.

TABLE III
COMPARISONS OF PREIMAGING WITH THE INTERPOLATION TECHNIQUES

	SSIM	PSNR	$\delta_{\frac{\pi}{8}}$	$\delta_{\frac{\pi}{4}}$	κ
Nearest	0.5064	21.8027	0.4015	0.6102	-1.1644
Bilinear	0.5874	23.0071	0.4378	0.6611	-0.6487
Cubic	0.6014	23.4050	0.4447	0.6681	-0.9944
Lanczos	0.6112	23.6173	0.4483	0.6726	-1.0468
ZP [a]	0.7824	26.2801	0.6129	0.8018	0.5029
$\uparrow$ [b]	0.2760	5.1974	0.2114	0.1916	—

[a]The imaging results after zero-padding $\mathbf{x}_0$.
[b]The maximum improvement comparisons with the competitors.

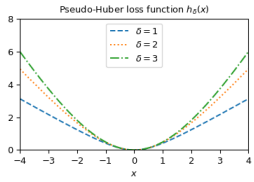


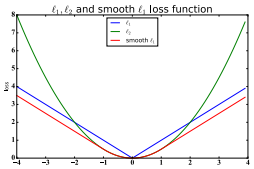
the radar image into the frequency domain and then fill the unknown components with zeros. The aligned image was obtained by re-imaging on the padded frequencies. To verify the preimaging technique, a set of experiments was performed. The proposed technique was compared with the typical interpolation techniques, including nearest, bilinear, cubic, and Lanczos tricks. The experimental results were tabulated in Table III. A set of exemplars were illustrated at the bottom.

As can be seen, all of the metrics obtained by the proposed method are much better than the linear interpolation techniques. Likewise, the phase congruence obtained by the proposed technique is much better than the interpretation skills. The metric $\delta_{(\pi/8)}$ for the reference method is around 0.6, compared to 0.8019 for the proposed method. So, the proposed technique could predict the phase more accurately. As for the fitting precision metric κ, all of the competitors provide negative values, compared to 0.5029 for the proposed method. The results demonstrate that the reference techniques do not fit the complex-valued SAR image well. This is because the SAR image formation mechanism is significantly different from the vision image. The general interpolation only considered the amplitude of the SAR image and ignored the imaging mechanism and the phase information.

TABLE IV
IMAGE METRICS OBTAINED BY DIFFERENT LOSS MEASUREMENTS

	MAE	MSE	Smoothℓ_1	Huber	pHuber	OurSR
SSIM	0.8134	0.8116	0.8134	0.8133	0.8137	0.8156
PSNR	27.96	27.87	27.96	27.95	27.96	28.12
$\delta_{\frac{\pi}{8}}$	0.6225	0.6199	0.6225	0.6225	0.6225	0.6241
$\delta_{\frac{\pi}{4}}$	0.8092	0.8076	0.8093	0.8093	0.8093	0.8111
κ	0.5477	0.5501	0.5457	0.5462	0.5476	0.5500
$\phi_{\mathcal{C}}$	4.7124	4.7681	4.7063	4.7326	4.7317	4.9203
$\phi_{\mathcal{E}}$	14.397	14.391	14.398	14.403	14.407	14.378





C. Verifications of Loss Measurements

For the task of image reconstruction, the representative functions to measure the regression losses include ℓ_1-loss, ℓ_2-loss, and the smooth ℓ_1-loss. They were used to implement the gradient descent algorithm. In this article, a combination of amplitude loss and phase loss was presented to measure the prediction error. The amplitude loss was quantized by the pseudo-Huber function, while the phase loss was calculated by the MAE function. To verify the presented training strategy, a set of experiments were performed. The proposed loss measurement strategy was compared to some of the representative loss functions. The experimental results are shown in Table IV.

From Table IV, we found that the learning effectiveness obtained by the proposed loss function is much better than the baseline functions. Four kinds of metrics are presented in this round of experiments, the accuracy of amplitude reconstruction and phase congruence, the fitting power of the model, and the information measurement. The performance obtained by the presented loss is consistently better than the reference functions. Take the amplitude accuracy, for example, the SSIM is 0.8156 for the proposed method and 0.0019 better than the nearest competitor (pHuber). Likewise, the image entropy for the proposed method is 14.378, 0.029 lower than pHuber. This is because both the amplitude and the phase measurements are considered simultaneously. Contrarily, the general functions only consider visual similarity. So, the super-resolution results produced by the proposed method are much better.

D. Ablation on Subaperture Fusion

This article presents a subaperture learning and hierarchical fusion method for complex-valued SAR image super-resolution. To study the effect of subaperture fusion, a set of ablation experiments was performed. The super-resolution via quarter subaperture fusion, half-aperture fusion, and

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TABLE V
ABLATION EXPERIMENTS FOR SUBAPERTURE FUSION

Method	SSIM	PSNR	$\delta_{\frac{\pi}{8}}$	$\delta_{\frac{\pi}{8}}$	κ
Preimaging	0.7824	26.28	0.6129	0.8018	0.5029
BaseAE	0.8107 0.0283↑	27.86 1.58↑	0.6206 0.0077↑	0.8083 0.0065↑	0.5241 0.0212↑
BaseAE(L1)	0.8128 0.21↑	28.04 0.0018↑	0.6208 0.0002↑	0.8085 0.0002↑	0.5370 0.0341↑
BaseAE(L12)	0.8141 0.13↑	28.05 0.0001↑	0.6220 0.0012↑	0.8094 0.0009↑	0.5397 0.0027↑
BaseAE(L123)	0.8156 0.15↑	28.12 0.0007↑	0.6241 0.0021↑	0.8111 0.0017↑	0.5500 0.0103↑

TABLE VI
MODEL EFFICIENCY OF THE PROPOSED METHOD

Method	BaseAE	BaseAE(L1)	BaseAE(L12)	BaseAE(L123)
FLOPs	1.3337E9	1.4097E9	3.5789E9	1.6000E10
MACs	2.2857E9	2.6184E9	1.1562E10	9.5221E10
Parameters	0.1207E6	0.1219E6	0.1815E6	0.3993E6

full-aperture fusion were pursued. The experimental results are shown in Table V.

1) "BaseAE," the raw encoder–decoder without subaperture fusion.
2) "BaseAE(L1)," the encoder–decoder with the quarter-aperture learning and fusion.
3) "BaseAE(L12)," the encoder–decoder with the quarter-aperture and half-aperture learning and fusion.
4) "BaseAE(L123)," the encoder–decoder with all three levels of subaperture learning and fusion.

From Table V, we found the subaperture learning and fusion strategy played a crucial role in the proposed method. In this article, three levels of learning and fusion were developed, including the quarter-aperture fusion, the half-aperture fusion, and the full-aperture fusion. The experimental results prove that each level of fusion could boost the reconstruction performance. The best performance can be obtained by considering all levels of subaperture fusion.

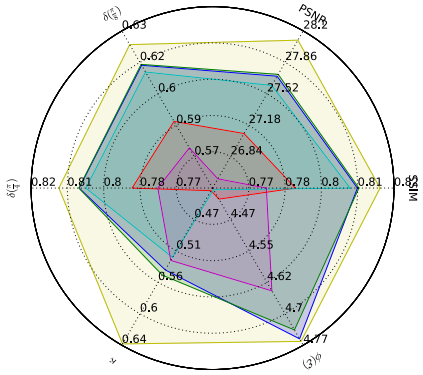
To further study the model efficiency of the presented method, some additional measurements were considered. Here, the floating-point operations per second (FLOPs), the multiply–accumulate operations (MACs), and the parameters were employed to quantize the computational complexity. The results were shown in Table VI. From Table VI, we found the computational complexity was gradually enhanced with the models increased. The FLOPs of three-level fusion were nearly ten times of BaseAE. Likewise, the MACs of three-level fusion were nearly 40 times of BaseAE. Yet, the computational cost was acceptable even with the common NVIDIA 3060 chip.

E. Comparisons With State-of-the-Art

In this article, a new complex-valued SAR image super-resolution method via a three-phase deep architecture was proposed. Different from the generic techniques, the radar imaging mechanism and the phase information were exploited simultaneously. To verify the proposed method, a set of experiments are performed. The proposed method is compared with several general vision super-resolution techniques. The baseline models include SRCNN [29] (with VGG backbone), EDSR, MDSR [30], RCAN [31], and DBPN [47]. The performance was assessed by the proposed evaluation system. The experimental results were displayed in Table VII, where the metrics were simultaneously visualized by a radar chart at the bottom.

TABLE VII
COMPARISONS WITH GENERAL SUPER-RESOLUTION ARCHITECTURE

	EDSR	MDSR	DBPN	SRVGG	RCAN	OurSR
SSIM	0.8079	0.8084	0.7874	0.7778	0.8055	0.8156
PSNR	27.71	27.73	27.09	26.60	27.61	28.11
$\delta_{\frac{\pi}{8}}$	0.6147	0.6151	0.5897	0.5779	0.6117	0.6241
$\delta_{\frac{\pi}{4}}$	0.8041	0.8042	0.7866	0.7781	0.8019	0.8111
κ	0.5394	0.5463	0.4335	0.5267	0.5227	0.5500
$\phi_{\mathcal{C}}$	4.7554	4.7340	4.4269	4.7052	4.6426	4.9203
$\phi_{\mathcal{E}}$	14.404	14.396	14.379	14.255	14.404	14.378



F. Verifications With Point Target Measurement

One of the most classical methods to measure radar image quality was using point target imaging. Different from the supervised evaluation strategy, there is no need for the reference image. So, the degraded images, the super-resolution result, and the full-resolution image can be assessed

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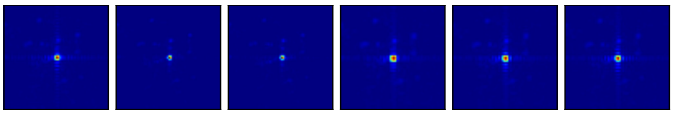
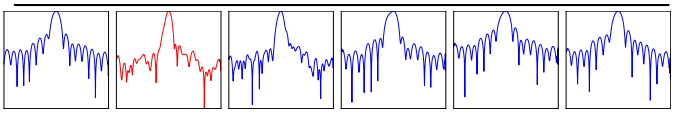


Fig. 22. From (left to right) 64 × 64-pixel patches obtained by zero-padding, super-resolution result, high-resolution image, bilinear, cubic, and Lanczos interpolation.

TABLE VIII
POINT TARGET IMAGING METRICS FOR AZIMUTH PROFILES

	LR	OurSR	HR	Bilinear	Cubic	Lanczos
PSLR	-16.464	-27.295	-26.577	-18.601	-13.706	-14.053
ISLR	-13.727	-24.783	-23.032	-16.886	-13.381	-12.937
IRW	126	87	89	166	149	140



individually. To verify the proposed method, a patch in 64 × 64 pixels around a peak is chosen from the original image. The patches obtained by zero-padding imaging, super-resolution result, fine-resolution image, and the interpolation results are displayed in Fig. 22. These patches were then interpolated in 8 times to 512 × 512 pixels by the sinc function

$$g(x) = \sum_i f(i) \cdot \mathrm{sinc}(x - i), \quad \mathrm{sinc}(x) = \frac{\sin(\pi x)}{\pi x} \tag{35}$$

where $f(x)$ is a band-limited signal and $g(x)$ is the interpolated signal. The azimuth and range profiles of interpolated images that passed through the peak were used to calculate PSLR, ISLR, and IRW. The unit of PSLR and ISLR is decibel (dB), while the unit of IRW is pixels. The details of the calculation can be found in Fig. 20.

1) Azimuth Profiles: The imaging metrics calculated by the azimuth profiles are shown in Table VIII, where the profiles from zero-padding imaging, super-resolution result, truth image, bilinear, cubic, and Lanczos interpolation were shown from left to right.

The imaging metrics obtained by the proposed method are consistently better than the reference methods. They are even better than the original full-resolution image. Take the azimuth profile, for example, the PSLR for the proposed method is −27.295 dB, compared to −26.577 and −16.464 dB for full-resolution image and the zero-padding imaging. The proposed method is 0.7180 and 10.8310 lower than them. The best performance obtained by the interpolation techniques is −18.601 (bilinear). Their performance is much poorer than the proposed method.

2) Range Profiles: The imaging metrics calculated by the range profiles were shown in Table IX, where the profiles resulting from zero-padding imaging, super-resolution result, full-resolution image, bilinear, cubic, and Lanczos interpolation are displayed from left to right.

The experimental results are similar to Table VIII. The metrics for the proposed method are consistently better than the reference methods. The PSLR, ISLR, and IRW for the proposed method are −29.783 dB, −20.039 dB, and 88-pixel, compared to −25.003 dB, −18.485 dB, and 88-pixel for the full-resolution image. Similarly, the performance of the interpolation techniques is much poorer than the proposed method.

TABLE IX
POINT TARGET IMAGING METRICS FOR RANGE PROFILES

	LR	OurSR	HR	Bilinear	Cubic	Lanczos
PSLR	-16.851	-29.783	-25.003	-18.240	-14.129	-14.818
ISLR	-13.257	-20.039	-18.485	-14.739	-12.247	-12.014
IRW	127	88	88	181	158	144

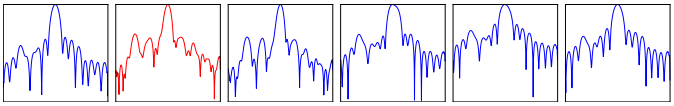
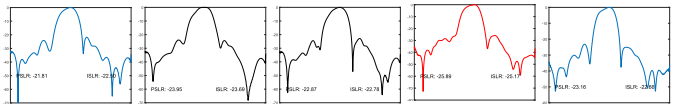


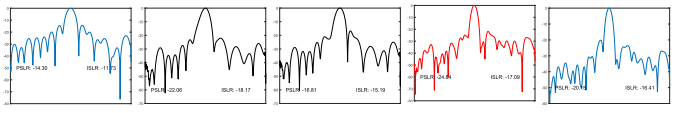
TABLE X
2-D POINT IMAGING MEASUREMENTS

2D	OurSR	HR	LR	Bilinear	Cubic	Lanczos
ISLR	-13.165	-12.466	-8.732	-10.515	-8.348	-8.021
↑		-0.699	-4.433	-2.650	-4.817	-5.144



LR Bilinear Cubic Proposed Full

	LR	Bilinear	Cubic	SR	Full
PSLR	-21.81	-23.95	-22.87	-25.89	-23.16
ISLR	-22.50	-23.69	-22.78	-25.17	-22.66



LR Bilinear Cubic Proposed Full

	LR	Bilinear	Cubic	Proposed	Full
PSLR	-14.30	-20.06	-16.81	-24.84	-20.15
ISLR	-11.73	-18.17	-15.19	-17.09	-16.41

Fig. 23. Measurements of point target: low-resolution image, bilinear interpretation result, cubic interpretation result, super-resolution image, and full-resolution image.

3) 2-D Metric: The 2-D ISLR of interpolated patches are shown in Table X, where ↑ denotes the improvement to the baseline methods. The metric for the proposed method is −13.165 dB. It is −0.699 dB, −4.433 dB, −2.650 dB, −4.817 dB, and −5.144 dB lower than the reference methods.

4) Additional Measurement: To demonstrate the image quality of the proposed method, another set of experiments was displayed. The results were displayed in Fig. 23. We found that the results were similar to the previous rounds of point targets. The linear interpolation techniques deteriorated the resolution. Likewise, the mainlobe has been widened simultaneously. The proposed method achieved the best performance. The ISLR,

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PSLR along the range, and the azimuth were much better than those competitors. The reconstructed result of the proposed method was more similar to the full-resolution image than the baseline. The good performance obtained by the proposed method can be attributed to the following issues.

1) The degradation process of the SAR image was learned and reproduced by a deep neural network. When the accumulation of historical data increases, the learning effectiveness can be improved continuously. The more the prior data, the better the learning effect. The richer the imaging scenarios, the better the reconstruction performance.
2) The zero padding technique was employed to form an initialized image. A model and data co-driven super-resolution paradigm was, therefore, formed. They promoted the super-resolution performance further.
3) Both the phase and the amplitude components were exploited in the proposed framework.

VI. Conclusion

A new complex-valued SAR image super-resolution via cross-domain learning and hierarchical fusion was proposed in this article. It could learn the degeneration process of SAR imaging and enhance the image resolution accordingly. On this foundation, an evaluation system for complex-valued SAR image super-resolution is established. Both the supervised and unsupervised evaluations are considered. Finally, multiple rounds of comparative experiments were performed. From the experimental results, some conclusions can be drawn.

1) Different from the general method, the amplitude and the phase of the SAR image are jointly exploited for SAR image super-resolution.
2) SAR imaging mechanism and deep learning are unified into a single framework, by which cross-domain learning and hierarchical fusion can be achieved.
3) The presented evaluation system is capable of measuring SAR image super-resolution comprehensively.

The results throw light on the superiority of the proposed method over the standard and recent techniques. The generated images with better quality could aid the downstream tasks.

In this article, a new subaperture learning and hierarchical fusion method was presented for SAR image super-resolution. One of the drawbacks is that the downstream interpretation task was not considered. The super-resolution reconstruction can be viewed as the intermediate process between SAR imaging and information perception. In the future, an intriguing study is to integrate the upstream super-resolution task with the downstream interpretation task and form a unified framework. It is expected that both the super-resolution performance and the image interpretation performance can be jointly boosted. We believe that the full potential of the joint learning thought is yet to be uncovered at present. In addition, how to combine the task of super-resolution and sidelobe suppression is a challenging problem [48].

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